Week 1: BASIC CONCEPTS IN ICT FOR LIBRARY AND INFORMATION SERVICES

INFORMATION

Meaning of Information

Information is **processed, organised, meaningful data** that reduces uncertainty in the mind of the recipient. It is anything that adds to a person's knowledge, helps decision-making, or solves a problem. Unlike raw data, information has been refined, interpreted, and given context.

In the library and information science (LIS) field, information is regarded as a **strategic resource**—just like money, human labour, or raw materials—because individuals and organisations depend on it to function effectively.

Characteristics of Information

1. **Accuracy** – Information must be correct and free from errors, because wrong information leads to wrong decisions.
2. **Relevance** – Information must meet the specific needs of the user. Irrelevant information is as good as useless.
3. **Timeliness** – Information must reach the user at the right time; old or delayed information may lose value.
4. **Completeness** – It should provide all necessary facts to support understanding.
5. **Clarity** – It must be easy to understand, avoiding ambiguity or confusion.
6. **Verifiability** – Reliable information can be checked or confirmed through other sources.
7. **Accessibility** – Information should be easy to obtain, especially in an ICT-rich environment.

Types of Information

* **Factual Information** – Raw, objective data (e.g., statistics, dates).

* **Analytical Information** – Information interpreted or analysed (e.g., research findings).

* **Current Information** – Up-to-date information needed in fields that change rapidly.

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* **Historical Information** – Past information that explains how things developed.

* **Reference Information** – Quick information used for facts (e.g., dictionaries, encyclopedias).

* **Scientific and Technical Information** – Generated through systematic research.

Importance of Information

* Supports learning and research.

* Aids decision-making in government, business, and education.

* Helps societal development and innovation.

* Enables communication and knowledge sharing.

* Drives economic activities and technological growth.

Information as a Resource

Modern societies operate on the concept of the **Information Economy** where information is treated as capital. Libraries exist largely to manage this resource by acquiring, organising, preserving, and disseminating information efficiently.

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2. COMMUNICATION

Meaning of Communication

Communication is the process of **transmitting information** from one person, group, or system to another through a medium. In simple terms, communication is the sharing of meaning.

In LIS, communication ensures that information passes effectively between librarians and users, staff and management, and the library and the outside world.

Elements of Communication

1. **Sender** – The originator of the message.
2. **Message** – The information or idea being communicated.
3. **Encoding** – Converting ideas into symbols (words, gestures, visuals).
4. **Medium/Channel** – The means of transmission (spoken word, email, radio, internet).
5. **Receiver** – The person for whom the message is intended.
6. **Decoding** – Interpreting or making meaning out of the message.
7. **Feedback** – The receiver's response, showing whether the message was understood.
8. **Noise** – Anything that distorts communication (e.g., poor network, wrong assumptions, language barrier).

Types of Communication

Oral Communication – Spoken words, announcements.

Written Communication – Reports, memos, emails.

Non-verbal Communication – Facial expressions, gestures.

Formal Communication – Official records, circulars.

Informal Communication – Casual conversations, chats.

Digital Communication – Social media, Zoom meetings, WhatsApp messages.

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Importance of Communication in Libraries

1. Facilitates smooth internal operations.
2. Encourages teamwork among library staff.
3. Improves reference and user-service delivery.
4. Enables user education and information literacy.
5. Helps libraries engage with their communities.

3. TECHNOLOGY

Meaning of Technology

Technology refers to the **tools, techniques, systems, and methods** humans use to solve problems and improve life.
In ICT-related courses, technology usually includes hardware, software, networks, and procedures that help in data processing and communication.

Forms of Technology Relevant to LIS

1. **Hard Technology** – Physical tools like computers, printers, scanners, servers, barcode readers.
2. **Soft Technology** – Software, applications, programming, cataloguing software.
3. **High Technology** – Advanced systems like artificial intelligence, robotics, cloud computing, Internet of Things.
4. **Assistive Technology** – Tools that help people with disabilities access information (screen readers, Braille terminals).

Roles of Technology in Libraries

* Enhances efficiency and speed of library operations.

* Reduces workload and human error.

* Enables digital libraries and automation.

* Improves access to global information through the internet.

* Supports remote access and self-service systems.

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4. INFORMATION TECHNOLOGY (IT)

Meaning of IT

Information Technology refers to all computer-based tools used for **creating, storing, processing, retrieving, and transmitting information**.
It focuses more on data handling and computer systems rather than communication networks.

Components of IT

* **Hardware** – Physical equipment.

* **Software** – Operating systems, application programs.

* **Databases** – Organised collections of information.

* **Procedures** – Operating instructions.

* **People** – IT professionals, users.

IT in Libraries

1. Used for cataloguing, classification, indexing.
2. Used for circulation control and acquisition.
3. Facilitates e-resources, OPAC, and digital repositories.
4. Supports storage of large volumes of data.

5. COMMUNICATION TECHNOLOGY

Meaning

Communication technology refers to digital systems used to **transmit information electronically** across distances. It focuses on connecting people and systems.

Examples include:

Email

Internet

Mobile phones

Social media

Video conferencing

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Telecommunication networks

Libraries rely heavily on communication technology for user engagement and digital service delivery.

6. INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

Meaning of ICT

ICT is the **combination of information technology and communication technology** to create, process, store, retrieve, and transmit information efficiently.
It is broader than IT because it includes both computing and telecommunication technologies.

Example:
A library computer system (IT) connected to the internet (communication technology) = ICT.

Components of ICT

Hardware – Computers, servers.

Software – Library management systems.

Networks – LAN, WAN, Wi-Fi.

Telecommunications – Internet service, modems, routers.

Data and Information – What is stored and transmitted.

People – Users and ICT staff.

Importance of ICT in Libraries

1. Enables library automation.
2. Supports online catalogues (OPAC).
3. Increases speed of reference services.

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4. Facilitates digital libraries and institutional repositories.
5. Enhances information literacy teaching.
6. Provides access to global electronic resources.
7. Supports remote users.

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